Banasthali Vidyapith
School of Automation
Course Handout: MCA IIIrd Semester/M.Sc. (DS) IIIrd Semester
B. Tech. (CS-AI/EC/EE/IT) VIIth Semester
July – December 2026

Date: 28 June 2026

Course Code: MCTR H426 **Course Name: 3D Printing and Designing**
Course Credits: 4 **Max. Marks: 100 (CA: 40 + ESA: 60)**
Course Instructor: Prof. Anshuman Shastri

Course Outcomes: The students will be able to:
CO 1 Understand the basic concepts of 3D printing
CO 2 Understand the utility difference of various 3D printing techniques
CO 3 Model a 3D design and analyse the structure
CO 4 Acquire the optimisation skills for designing and fabrication
CO 5 Analyse a fabricated design to optimise a design model
CO 6 Learn about the maintenance and troubleshooting of a printing system

Syllabus

Section A

Introduction 3-Dimensional printing: 3D Printing vs traditional techniques, Advantages of 3D Printing, Basic Introduction to solid modelling, designing, Introduction to STL file, slicing and layering.
Costs of 3D printers, Cost estimation of 3DP, Introduction to industrial, educational and research, household and personal 3D Printers, Introduction of open-source printers, building and developing 3D printer for custom applications.
3D printing technologies: Fused filament fabrication (FFF)- Fused Deposition Modelling (FDM), Stereolithography (SLA), Selective Laser Melting (SLM), Binder Jetting, Selective Laser Sintering (SLS), Inkjet Printing: Dimatix and household, Aerosol jet printing, Polyjet printing, Digital Light Processing (DLP)

Section B

Modelling in 3-Dimensions: Software basic principles, boundary conditions, machine limitations
Modelling software and their features: AutoCAD, 123D, introduction to other freely available software, Point cloud to STL Conversion.
Pre-processing: Data Preparation, STL error removal, Printing technique identification, Printing parameter selection, Build orientation, Material selection, Printing system limitations, Cost-analysis.
CAD model optimization: Introduction to design optimisation, requirements of optimisation, CAD model preparation for 3D Printing, CAD model modification for printer limitation.
Model infill: Requirement, Necessity, types, realisation. Printer calibration
Slicing of CAD model: Type of slicing, Slicing algorithms
Analysis of a failed model for design corrections, Troubleshooting a printer system, Introduction to Post-processing requirements
Maintaining a 3D printer: System maintenance, extruder maintenance, motor maintenance, new and old material maintenance, pressure system maintenance.

Section C

Advanced 3D Modelling: Modelling complex shapes and structures, software, and system tolerances.
Customising an advanced design print: build plate adhesion, infill adjustments, print speed modifications, corner speed alterations, support structures.
Post-processing requirements: Cleaning, support structure removal, sanding, adhesion, mounting, curing, heating, smoothing, polishing, and electroplating.
Industrial printer systems: Polymer-based printers, metal-based printers, powder-based printers, a brief introduction to other printers.
Multi-extrusion systems: printing with multiple extruders, printing with multiple materials, printing with metals, Latest advancements in industrial 3D printing.
Applications and case studies: Engineering applications, medical applications, Reverse Engineering.

Suggested Books:

R1. Pham, D.T., Demov, S.S., Rapid Manufacturing: The Technologies and Applications of Rapid Prototyping and Rapid Tooling, Springer-Verlag London Limited, 2001.

R2. Hopkinson, N., Hague, R.J.M. and Dickens, P.M., Rapid Manufacturing and Industrial Revolution for the Digital Age, John Wiley and Sons Ltd, Chichester, 2005.

R3. Noorani, R., Rapid Prototyping: Principles and Applications, John Wiley & Sons, Inc., New Jersey, 2006.

R4. Zeid, I., Mastering CAD/CAM, Tata McGraw Hill, 2006

R5. Gebhardt, A., Rapid Prototyping, Hanser Gardner Publications, Inc., Cincinnati, 2003.

R6. Gibson, I., Software Solutions for Rapid Prototyping, Professional Engineering Publication Ltd., 2002.

R7. Patri, K. V., and Weiyin, Ma, Rapid Prototyping - Laser-based and Other Technologies, Kluwer Academic Publishers, U.S.A., 2003.

Suggested E-Learning Material:

E1. Subject guides: 3D printing in the Mardigian Library: Resources for Faculty. Resources for Faculty - 3D Printing in the Mardigian Library - Subject Guides at University of Michigan-Dearborn. (n.d.). Retrieved November 5, 2021, from https://guides.umd.umich.edu/c.php?g=558736&p=3935287

E2. Library guides: 3D printing: 3D model sources and ideas. 3D Model Sources and Ideas - 3D Printing - Library Guides at Penn State University. (n.d.). Retrieved November 5, 2021, from https://guides.libraries.psu.edu/3d/models

E3. https://nptel.ac.in/courses/112/104/112104265/

E4. 3D printing and Additive Manufacturing: Vol 0, no 0. (n.d.). Retrieved November 5, 2021, from https://www.liebertpub.com/toc/3dp/0/0

Assessment Schedule:

Component	Marks	Submission/ Examination Date	Allotment
Home assignment I	10	27 August, 2026**	Topics shall be allotted in the class by 5 August 2026
Periodical test I	10	9-12 September, 2026*	-
Home assignment II	10	7 October, 2026**	Topics shall be allotted in the class by 14 September, 2026
Periodical test II	10	30 October – 2 November, 2026*	-
Semester Examination	60	2-19 December, 2026*	-

* Subject to change.
**Assignment marks will be based on written documents, viva-voce or/ and any other component(s) decided by the instructor on regular basis.

Lecture-Wise Schedule:

Lecture Number	Topics to be covered	Text/Ref. Books
	Section A	
1-4	Introduction to 3D Printing, Advancements in 3D Printing, Advantages of 3D Printing	E1
5-9	Overview of types of 3D printing: Similarities and differences, 3D Printing vs traditional techniques	R1, R2, R3
10-13	Fundamentals and functioning of 3D Printing, Introduction to 3D modelling	R2, R3, R4
14-23	Types of 3D printers: Filament based 3D printers, powder based 3d printers, resin-based 3D printers, conductive ink-based 3D Printers, UV lighted based 3D Printers, ceramic based 3D printers. Printing material: filaments, polymer, metals, ceramics, powders, others.	R1, R2, E2, E3
24-25	3D modelling tools, slicing, G-codes, M-codes, STL files, layering	R4, R6
	Section B	
26-27	3D Modelling: Boundary conditions, software limitations, printer limitations	R4, R6
28-30	3D Modelling software: AutoCAD, TinkerCAD, 123D, other freely available software	R4, R6
31-34	Pre-processing: Data Preparation, STL error removal, Printing technique identification, Printing parameter selection, Build orientation, Material selection, Printing system limitations, Cost-analysis. Slicing of CAD model: Type of slicing, Slicing algorithm	R2, R3
35-37	CAD model optimization: Introduction to design optimisation, requirements of optimisation, CAD model preparation for 3DP, Parametric 3D modelling, CAD model modification for printer limitation. Analysis of a failed model for design corrections Customising an advanced design print: build plate adhesion, infill adjustments, print speed modifications, corner speed alterations, support structures. Post-processing requirements: Cleaning, support structure removal, sanding, adhesion, mounting, curing, heating, smoothing, polishing, and electroplating.	R2, R3, R6, E1, E2 R3, R4
	Section C	
38-40	Troubleshooting a printer system Introduction to Post-processing requirements Maintaining a 3D printer: System maintenance, extruder maintenance, motor maintenance, new and old material maintenance, pressure system maintenance.	R3, R4
41-43	Industrial printer systems: Polymer-based printers, metal-based printers, a brief introduction to other printers. Multi-extrusion systems: printing with multiple polymers, printing with multiple materials, printing with metals. Latest advancements in industrial 3D printing.	R1, R2, R3, R7, E3, E4
44-45	Applications and case studies: Engineering applications, medical applications, Reverse Engineering.	R2, R3, R4
	Further case studies, 3D printer advancement discussions and revision of topics as per students' request	

Prof. Anshuman Shastri